

Homework 5

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4.2.1

(1) The eigenvalues of L_z : $-1, 0, 1$.

(2) The eigenvector of L_z corresponding to $l_z = 1$ is $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

$$\langle L_x \rangle = (1 \ 0 \ 0) \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$\langle L_x^2 \rangle = (1 \ 0 \ 0) \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2}$$

$$(\Delta L_x)^2 = \langle (L_x - \langle L_x \rangle)^2 \rangle = \langle L_x^2 \rangle - \langle L_x \rangle^2 = \frac{1}{2} - 0 = \frac{1}{2}$$

$$\Delta L_x = \frac{1}{\sqrt{2}}$$

(3)

$$\det(L_x - \lambda I) = \begin{vmatrix} -\lambda & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\lambda & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & -\lambda \end{vmatrix} = -\lambda^3 + \lambda = 0$$

$$\lambda = 0, \pm 1$$

$$l_x = -1 : \begin{pmatrix} 1 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 1 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad |l_x = -1\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}$$

$$l_x = 0 : \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad |l_x = 0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$l_x = 1 : \begin{pmatrix} -1 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -1 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad |l_x = 1\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}$$

(4) The eigenvector of L_z corresponding to $l_z = -1$ is $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

$$|\langle l_x = -1 | l_z = -1 \rangle|^2 = \left| \frac{1}{2} (1 \ -\sqrt{2} \ 1) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right|^2 = \left| \frac{1}{2} \right|^2 = \frac{1}{4}$$

$\Rightarrow$ The probability of the outcome $l_x = -1$ is $\frac{1}{4}$.

$$|\langle l_x = 0 | l_z = -1 \rangle|^2 = \left| \frac{1}{\sqrt{2}} (1 \ 0 \ -1) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right|^2 = \left| -\frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

$\Rightarrow$ The probability of the outcome $l_x = 0$ is $\frac{1}{2}$.

$$|\langle l_x = 1 | l_z = -1 \rangle|^2 = \left| \frac{1}{2} (1 \sqrt{2} 1) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right|^2 = \left| \frac{1}{2} \right|^2 = \frac{1}{4}$$

$\Rightarrow$ The probability of the outcome $l_x = 1$ is $\frac{1}{4}$.

In fact,

$$\begin{aligned} |l_z = -1\rangle &= |l_x = -1\rangle \langle l_x = -1 | l_z = -1\rangle + |l_x = 0\rangle \langle l_x = 0 | l_z = -1\rangle + \\ &\quad |l_x = 1\rangle \langle l_x = 1 | l_z = -1\rangle \\ &= \frac{1}{2} |l_x = -1\rangle - \frac{1}{\sqrt{2}} |l_x = 0\rangle + \frac{1}{2} |l_x = 1\rangle. \end{aligned}$$

$$(5) \quad |\psi\rangle = C_{-1} |l_z = -1\rangle + C_0 |l_z = 0\rangle + C_1 |l_z = 1\rangle$$

$$C_{-1} = \langle l_z = -1 | \psi \rangle = (0 \ 0 \ 1) \begin{pmatrix} \frac{1}{2} \\ \frac{1}{\sqrt{2}} \\ \frac{1}{2} \end{pmatrix} = \frac{1}{\sqrt{2}}$$

$$C_0 = \langle l_z = 0 | \psi \rangle = (0 \ 1 \ 0) \begin{pmatrix} \frac{1}{2} \\ \frac{1}{\sqrt{2}} \\ \frac{1}{2} \end{pmatrix} = \frac{1}{2}$$

$$C_1 = \langle l_z = 1 | \psi \rangle = (1 \ 0 \ 0) \begin{pmatrix} \frac{1}{2} \\ \frac{1}{\sqrt{2}} \\ \frac{1}{2} \end{pmatrix} = \frac{1}{2}$$

$$\Rightarrow |\psi\rangle = \frac{1}{\sqrt{2}} |l_z = -1\rangle + \frac{1}{2} |l_z = 0\rangle + \frac{1}{2} |l_z = 1\rangle$$

Since L_z^2 is measured to be $+1$, l_z can be 1 or -1 .

The probability of this result is: $|C_{-1}|^2 + |C_1|^2 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$.

After the measurement, the state is

$$|\psi'\rangle = C'_1 |l_z = -1\rangle + C'_1 |l_z = 1\rangle$$

$$C'_1 / C_{-1} = C_1 / C_{-1} = \frac{1}{\sqrt{2}}$$

$$\text{Normalization: } C_1'^2 + C_{-1}'^2 = 1 \Rightarrow C'_1 = \sqrt{\frac{1}{3}}, \quad C_{-1}' = \sqrt{\frac{2}{3}}$$

$$|\psi'\rangle = \sqrt{\frac{2}{3}} |l_z = -1\rangle + \sqrt{\frac{1}{3}} |l_z = 1\rangle.$$

If we measure L_z ,

outcome $l_z = -1$, probability $|C_{-1}'|^2 = \frac{2}{3}$;

outcome $l_z = 1$, probability $|C_1'|^2 = \frac{1}{3}$.

$$(6) \quad |\psi\rangle = C_1 |L_z = 1\rangle + C_0 |L_z = 0\rangle + C_{-1} |L_z = -1\rangle$$

$$|C_1|^2 = P(L_z = 1) = \frac{1}{4} \Rightarrow C_1 = \frac{1}{2} e^{i\delta_1}$$

$$|C_0|^2 = P(L_z = 0) = \frac{1}{2} \Rightarrow C_0 = \frac{1}{\sqrt{2}} e^{i\delta_2}$$

$$|C_{-1}|^2 = P(L_z = -1) = \frac{1}{4} \Rightarrow C_{-1} = \frac{1}{2} e^{i\delta_3}$$

Therefore,

$$|\psi\rangle = \frac{1}{2} e^{i\delta_1} |L_z = 1\rangle + \frac{1}{\sqrt{2}} e^{i\delta_2} |L_z = 0\rangle + \frac{1}{2} e^{i\delta_3} |L_z = -1\rangle.$$

$\delta_1, \delta_2, \delta_3$ cannot be determined individually.

It is true that $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ represent the same state, in the sense that θ does not change any physical observable. However, if $|\psi\rangle$ is a linear combination of several eigenstates, as the case shown above, the relative values between the phase factors in front of each eigenstates, $e^{i\delta_1}$, $e^{i\delta_2}$ and $e^{i\delta_3}$ do have physical significance.

For example,

$$P(L_x=0) = |\langle L_x=0 | \psi \rangle|^2$$

$$= \left| \frac{1}{2} e^{i\delta_1} \langle L_x=0 | L_z=1 \rangle + \frac{1}{\sqrt{2}} e^{i\delta_2} \langle L_x=0 | L_z=2 \rangle + \frac{1}{2} e^{i\delta_3} \langle L_x=0 | L_z=-1 \rangle \right|^2$$

$$\langle L_x=0 | L_z=1 \rangle = \frac{1}{\sqrt{2}} (1 \ 0 \ -1) \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}$$

$$\langle L_x=0 | L_z=0 \rangle = \frac{1}{\sqrt{2}} (1 \ 0 \ -1) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$\langle L_x=0 | L_z=-1 \rangle = \frac{1}{\sqrt{2}} (1 \ 0 \ -1) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = -\frac{1}{\sqrt{2}}$$

$$\Rightarrow P(L_x=0) = \left| \frac{1}{2} \cdot \frac{1}{\sqrt{2}} e^{i\delta_1} - \frac{1}{2} \cdot \frac{1}{\sqrt{2}} e^{i\delta_3} \right|^2$$

$$= \frac{1}{8} (e^{-i\delta_1} - e^{-i\delta_3})(e^{i\delta_1} - e^{i\delta_3})$$

$$= \frac{1}{8} [2 - (e^{-i(\delta_1-\delta_3)} + e^{i(\delta_1-\delta_3)})]$$

$$= \frac{1}{8} \left[2 - 2 \cdot \frac{e^{i(\delta_1-\delta_3)} + e^{-i(\delta_1-\delta_3)}}{2} \right]$$

$$= \frac{1}{4} [1 - \cos(\delta_1 - \delta_3)]$$

Therefore, the difference between δ_1 and δ_3 can be measured by physical experiments.

4.2.2

Since $\psi(x)$ is real, $\psi^*(x) = \psi(x)$

$$\begin{aligned} \langle p \rangle &= \int_{-\infty}^{\infty} \psi^*(x) (-i\hbar \frac{\partial}{\partial x}) \psi(x) dx \\ &= -i\hbar \int_{-\infty}^{\infty} \psi(x) \frac{\partial}{\partial x} \psi(x) dx \\ &= -\frac{1}{2} i\hbar \int_{-\infty}^{\infty} \frac{\partial}{\partial x} \psi^2(x) dx \\ &= -\frac{1}{2} i\hbar \psi^2(x) \Big|_{-\infty}^{\infty} \quad * \quad \psi(x) \rightarrow 0 \text{ as } x \rightarrow \pm\infty \\ &= 0 \end{aligned}$$

For general case, $\langle p \rangle = \int_{-\infty}^{\infty} [C\psi(x)]^* (-i\hbar \frac{\partial}{\partial x}) [C\psi(x)] dx = |C|^2 \int_{-\infty}^{\infty} \psi(x) \frac{\partial}{\partial x} \psi(x) dx$

$$= |C|^2 \cdot 0 = 0$$

4.2.3

$$\langle p \rangle = -i\hbar \int_{-\infty}^{\infty} \psi^*(x) \frac{\partial}{\partial x} \psi(x) dx$$

$$\begin{aligned} \langle p' \rangle &= -i\hbar \int_{-\infty}^{\infty} [\psi^*(x) e^{-ip_0 x/\hbar}] \frac{\partial}{\partial x} [\psi(x) e^{ip_0 x/\hbar}] dx \\ &= -i\hbar \int_{-\infty}^{\infty} [\psi^*(x) e^{-ip_0 x/\hbar}] \left[\frac{\partial}{\partial x} \psi(x) \right] \cdot e^{ip_0 x/\hbar} dx \\ &\quad - i\hbar \int_{-\infty}^{\infty} [\psi^*(x) e^{-ip_0 x/\hbar}] \cdot \psi(x) \cdot i p_0 / \hbar \cdot e^{ip_0 x/\hbar} dx \\ &= -i\hbar \int_{-\infty}^{\infty} \psi^*(x) \frac{\partial}{\partial x} \psi(x) dx - i\hbar \cdot i p_0 / \hbar \cdot \int_{-\infty}^{\infty} \psi^*(x) \psi(x) dx \\ &= \langle p \rangle + p_0 \int_{-\infty}^{\infty} \psi^*(x) \psi(x) dx \\ &= \langle p \rangle + p_0 \end{aligned}$$

Problem 3.

$$\begin{aligned} \langle x \rangle &= \langle \psi | x | \psi \rangle \\ &= \iint dp dp' \langle \psi | p \rangle \langle p | x | p' \rangle \langle p' | \psi \rangle \\ &= \iint dp dp' \psi^*(p) \langle p | x | p' \rangle \psi(p') \\ &= \iint dp dp' \psi^*(p) i\hbar \frac{\partial}{\partial p} \delta(p - p') \psi(p') \\ &= \int \psi^*(p) \left(i\hbar \frac{\partial}{\partial p} \right) \psi(p) dp \end{aligned}$$

Where we use the relation $\langle p' | x | p'' \rangle = i\hbar \frac{\partial}{\partial p'} \delta(p' - p'')$ Prove: $\langle p' | x | p'' \rangle$

$$\begin{aligned} &= \iint dx' dx'' \langle p' | x' \rangle \langle x' | x | x'' \rangle \langle x'' | p'' \rangle \\ &= \iint dx' dx'' \frac{1}{\sqrt{2\pi\hbar}} e^{-ip'x'/\hbar} \cdot x'' \delta(x' - x'') \cdot \frac{1}{\sqrt{2\pi\hbar}} e^{ip''x''/\hbar} \\ &= \int dx' \frac{1}{2\pi\hbar} x' e^{-i(p' - p'')x'/\hbar} \\ &= \frac{1}{2\pi\hbar} \int dx' \left(-\frac{\hbar}{i} \right) \frac{\partial}{\partial p'} e^{-i(p' - p'')x'/\hbar} \\ &= \frac{1}{2\pi\hbar} (i\hbar) \frac{\partial}{\partial p'} \int dx' e^{-i(p' - p'')x'/\hbar} \\ &= \frac{1}{2\pi\hbar} (i\hbar) \frac{\partial}{\partial p'} [2\pi\hbar \delta(p' - p'')] \\ &= i\hbar \frac{\partial}{\partial p'} \delta(p' - p'') \end{aligned}$$

Problem 4

In the energy eigenbasis, $\hat{H} = \begin{pmatrix} E_1 & & \\ & E_2 & \\ & & \ddots \end{pmatrix}$

$$\begin{aligned} \text{Therefore, } e^{-i\hat{H}t/\hbar} &= \sum_{n=0}^{\infty} \frac{1}{n!} (-i\hat{H}t/\hbar)^n \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \left[-i \begin{pmatrix} E_1 & & \\ & E_2 & \\ & & \ddots \end{pmatrix} t/\hbar \right]^n \\ &= \begin{pmatrix} \sum_{n=0}^{\infty} \frac{1}{n!} (-iE_1t/\hbar)^n & & \\ & \sum_{n=0}^{\infty} \frac{1}{n!} (-iE_2t/\hbar)^n & \\ & & \ddots \end{pmatrix} \\ &= \begin{pmatrix} e^{-iE_1t/\hbar} & & \\ & e^{-iE_2t/\hbar} & \\ & & \ddots \end{pmatrix} \\ &= \sum_E |E\rangle\langle E| e^{-iEt/\hbar} \end{aligned}$$

$$\text{Thus, } U(t) = \sum_E |E\rangle\langle E| e^{-iEt/\hbar} = e^{-i\hat{H}t/\hbar}$$

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle &= i\hbar \frac{\partial}{\partial t} (e^{-i\hat{H}t/\hbar} |\psi(0)\rangle) \\ &= i\hbar (-i\hat{H}/\hbar) e^{-i\hat{H}t/\hbar} |\psi(0)\rangle \\ &= \hat{H} (e^{-i\hat{H}t/\hbar} |\psi(0)\rangle) \\ &= \hat{H} |\psi(t)\rangle \end{aligned}$$

Therefore, $|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(0)\rangle$ satisfies the Schrödinger equation.

Problem 5

$$|\psi\rangle = a|\psi_1(t)\rangle + b|\psi_2(t)\rangle$$

$$= a|\psi_1\rangle e^{-iE_1t/\hbar} + b|\psi_2\rangle e^{-iE_2t/\hbar}$$

$$P(x) = |\langle x|\psi\rangle|^2 = (a^* e^{iE_1t/\hbar} \langle\psi_1|x\rangle + b^* e^{iE_2t/\hbar} \langle\psi_2|x\rangle) \cdot$$

$$(a e^{-iE_1t/\hbar} \langle x|\psi_1\rangle + b e^{-iE_2t/\hbar} \langle x|\psi_2\rangle)$$

$$= |a\psi_1(x)|^2 + |b\psi_2(x)|^2 + a^*b\psi_1^*(x)\psi_2(x) e^{i(E_1-E_2)t/\hbar} + ab^*\psi_1(x)\psi_2^*(x) e^{i(E_2-E_1)t}$$

Therefore, the probability of finding the particle at some position changes with time.

Problem 2 (Done by Sean Finch)

Make Plots of the probability density in space and momentum space for a Gaussian wavepacket for several values of Delta to illustrate the X and P uncertainty relation.

Note: here we set $a = 0$ (the wavepacket centers at $x=0$) and $\hbar = 1$ (makes the scaling of the graphs easier to read).

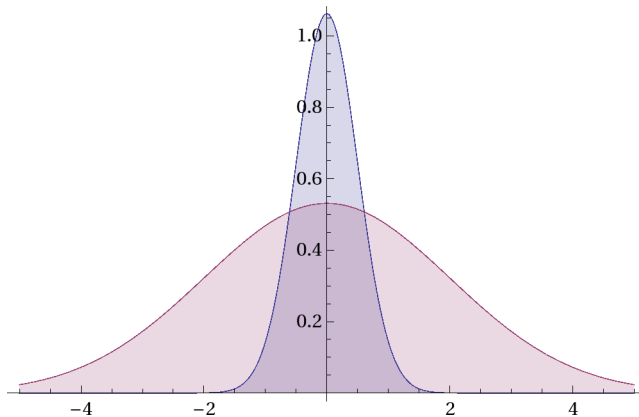
Definitions of the wave packet functions:

$$X[x_, d_] := 1 / (\text{Pi} * d^2)^{.25} * \text{Exp}[-x^2 / (2 d^2)]$$

$$P[p_, d_] := (d^2 / \text{Pi})^{.25} * \text{Exp}[-(p * d)^2 / (2)]$$

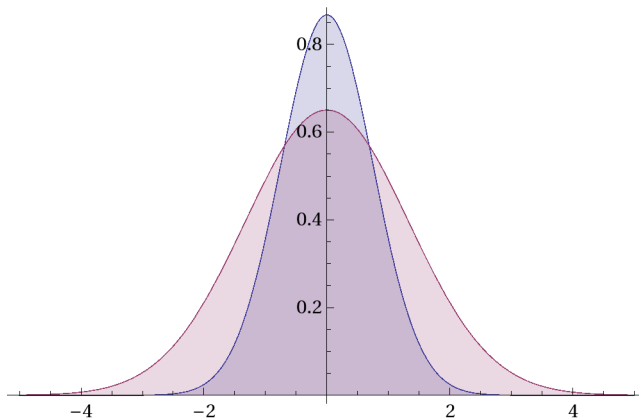
Here is a plot with $\Delta = .5$: (this means the position is more localized)

```
Plot[{X[x, .5], P[x, .5]}, {x, -5, 5}, Filling -> Axis]
```



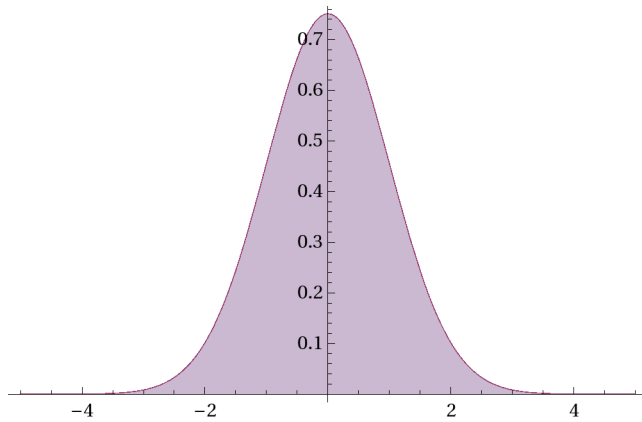
$\Delta = .75$: (position is more localized)

```
Plot[{X[x, .75], P[x, .75]}, {x, -5, 5}, Filling -> Axis]
```



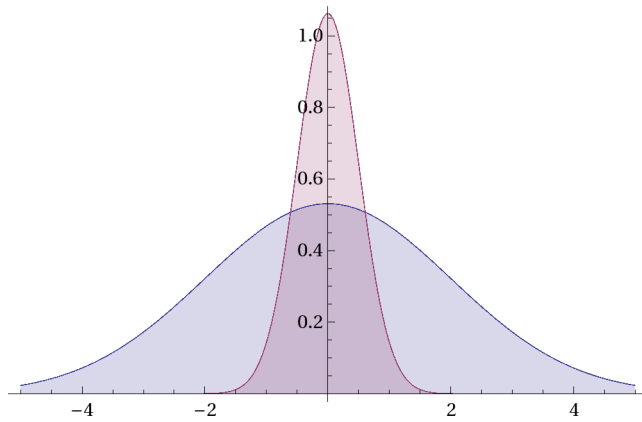
$\Delta = 1$ (X and P know to the same amount of precision)

```
Plot[{X[x, 1], P[x, 1]}, {x, -5, 5}, Filling → Axis]
```



Delta = 2: (momentum is more localized)

```
Plot[{X[x, 2], P[x, 2]}, {x, -5, 5}, Filling → Axis]
```



Delta = 2: (momentum in very small range)

```
Plot[{X[x, 5], P[x, 5]}, {x, -10, 10},  
PlotRange → Full, Filling → Axis]
```

